

flights	
departure_a...	INTEGER
arrival_airpo...	INTEGER
airline_id	INTEGER
flight_id	INTEGER
created_at	DATE
updated_at	DATE
departure_gate	TEXT
arrival_gate	TEXT
sched_departure_ti...	TIME
sched_arrival_time	TIME
actual_departure_t...	TIME
actual_arrival_time	TIME

airlines	
airline_code	TEXT
name	TEXT
country	TEXT
airline_id	INTEGER
created_at	DATE
updated_at	DATE

passengers	
first_name	TEXT
last_name	TIME
gender	TEXT
birth_date	DATE
citizenship	TEXT
residence	TEXT
passenger_id	INTEGER
passport_nu...	INTEGER
created_at	DATE
updated_at	DATE

airport	
airport_id	INTEGER
name	TEXT
country	TEXT
state	TEXT
city	TEXT
created_at	DATE
updated_at	DATE

booking	
flight_id	INTEGER
booking_id	INTEGER
platform	TEXT
passenger_id	INTEGER
created_at	DATE
updated_at	DATE
price	INTEGER
status	TEXT

security check	
check_id	INTEGER
passenger_id	INTEGER
result	TEXT
created_at	DATE
updated_at	DATE

booking changes	
booking_id	INTEGER
change_id	INTEGER
created_at	DATE
updated_at	DATE

baggage	
baggage_id	INTEGER
booking_id	INTEGER
weight	TEXT
created_at	DATE
updated_at	DATE

boarding	
pass_id	INTEGER
booking_id	INTEGER
seat	INTEGER
board_time	TIME
created_at	DATE
updated_at	DATE

baggage checking	
checking_id	INTEGER
booking_id	INTEGER
passenger_id	INTEGER
result	TEXT
created_at	DATE
updated_at	DATE

Laboratory Work 1 - ERD Documentation

System: International Airport Database

The database is designed to manage core airport operations, including airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checking, booking changes, and passenger security checks. It stores scheduled and actual flight information and links operational records to the relevant passenger, booking, flight, airline, and airport.

Entity Descriptions

Entity	Description
Airport	Stores airport information such as identifier, name, country, state, city, and record timestamps.
Airlines	Stores airline company information, including identifier, airline code, name, country, and record timestamps.
Flights	Stores flight information, including airline, departure and arrival airports, gates, scheduled times, actual times, and record timestamps.
Passengers	Stores passenger details such as name, gender, birth date, citizenship, residence, passport number, and record timestamps.
Booking	Stores a passenger flight booking, including booking identifier, flight, passenger, status, platform, price, and record timestamps.
Booking Changes	Stores information about changes made to a booking.
Baggage	Stores baggage registered for a booking, including baggage identifier, booking, weight, and record timestamps.
Baggage Checking	Stores baggage-check results linked to the relevant booking and passenger.
Boarding	Stores boarding-pass information such as pass identifier, booking, seat, boarding time, and record timestamps.
Security Check	Stores airport security-check results for a passenger.

Relationship and Participation Summary

- Each flight is associated with an airline and with departure and arrival airports.
- Each booking is linked to a flight and to a passenger.
- Booking-change records are created only when a booking is changed, so this participation is optional for a booking.
- Baggage registration is optional because passengers may travel without baggage; registered baggage belongs to a booking.
- Baggage-check records store the check result for registered baggage and are linked to the booking and passenger.
- Boarding-pass information is linked to a booking.
- Each passenger passes airport security, and the security-check record is linked to that passenger.

Legend / Notation

Notation	Meaning
Entity box	Represents a table/entity in the database.
Attribute row	Represents an attribute/column belonging to an entity.
Key symbol	Marks a key attribute used to identify or reference records.
Connector line	Shows a relationship between two entities.
1	Represents the "one" side of a relationship.
n	Represents the "many" side of a relationship.